

Resolucao: 28x28  
Retreino dos pesos: SGD

Learning rate: n/a

## Validacao visual das predicoes - KMNIST

Variantes selecionadas: hyperneat

Imagem original, mapas de oclusao e diferenca entre NEAT puro e NEAT + SGD.

Baseline + SGD: 73.2%

NEAT puro: 15.2%

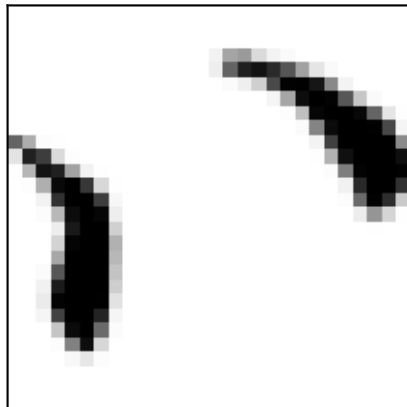
Topologia NEAT + SGD: 62.0%

Conexoes ativas: 6

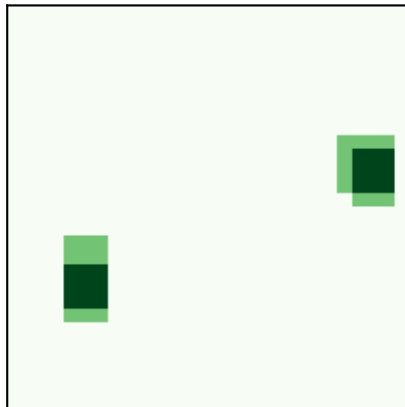
| Variante                        | NEAT puro | NEAT + SGD | Fitness | Conexoes |
|---------------------------------|-----------|------------|---------|----------|
| neat_mnist_14x14                | 23.4%     | 42.0%      | 0.679   | 442      |
| neat_mnist_14x14_sparse         | 25.0%     | 35.8%      | 0.698   | 135      |
| neat_mnist_14x14_dense          | 24.8%     | 42.4%      | 0.642   | 811      |
| neat_mnist_14x14_hidden         | 26.4%     | 40.0%      | 0.661   | 522      |
| neat_mnist_features_full        | 27.8%     | 39.8%      | 0.721   | 347      |
| neat_mnist_prototypes_full      | 22.2%     | 36.2%      | 0.639   | 182      |
| neat_mnist_seeded_prototypes    | 48.6%     | 47.8%      | 1.466   | 200      |
| hyperneat_mnist_cppn            | 15.2%     | 62.0%      | 0.469   | 6        |
| hyperneat_mnist_features_cppn   | 21.0%     | 61.2%      | 0.498   | 5        |
| hyperneat_mnist_prototypes_cppn | 14.6%     | 62.0%      | 0.475   | 5        |

Baseline: 5 (100.0%) | NEAT puro: 3 (10.0%) | NEAT + SGD: 5 (90.7%) | similaridade NEAT/SGD: 0.41 | pixels comuns: 46.2%

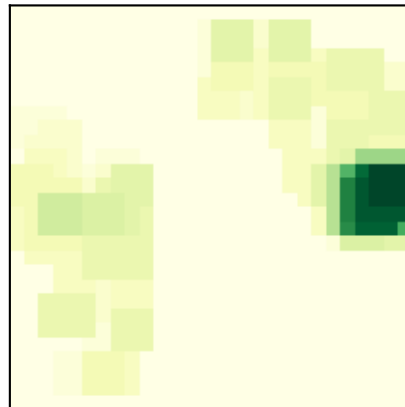
Imagem original



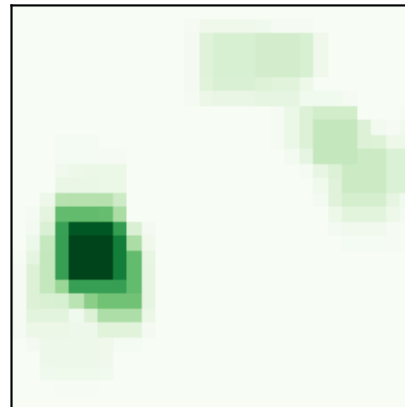
Olhar: baseline



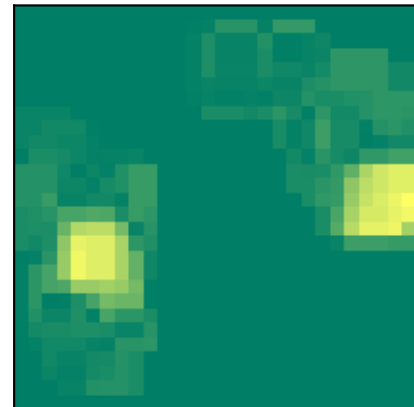
Olhar: NEAT puro



Olhar: NEAT + SGD

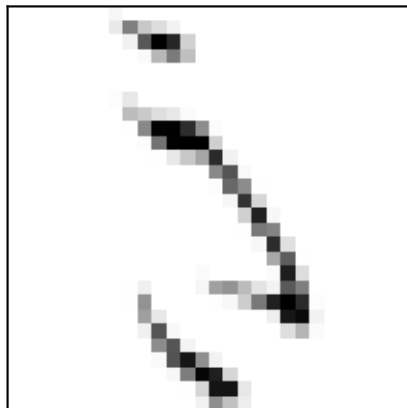


Diferenca NEAT vs SGD

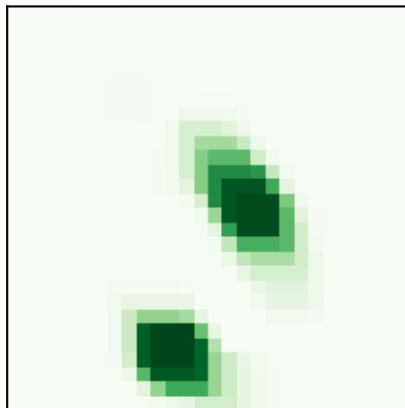


Baseline: 1 (98.7%) | NEAT puro: 9 (10.0%) | NEAT + SGD: 1 (78.1%) | similaridade NEAT/SGD: 0.78 | pixels comuns: 61.5%

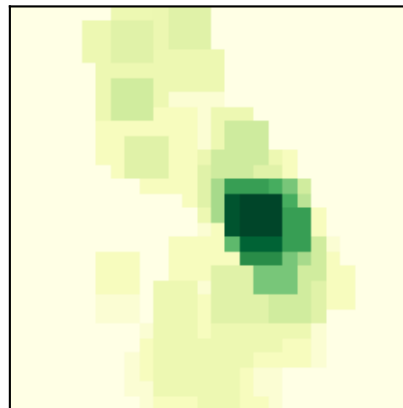
Imagem original



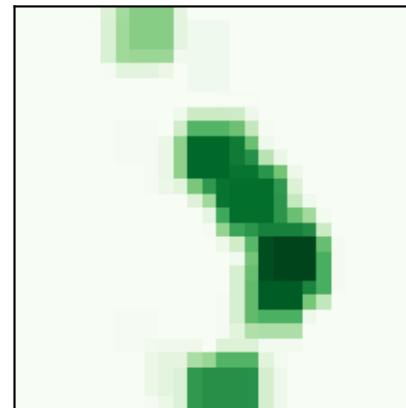
Olhar: baseline



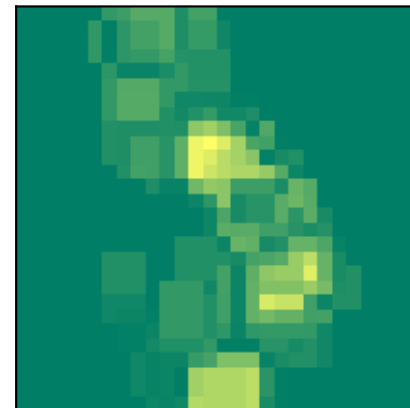
Olhar: NEAT puro



Olhar: NEAT + SGD

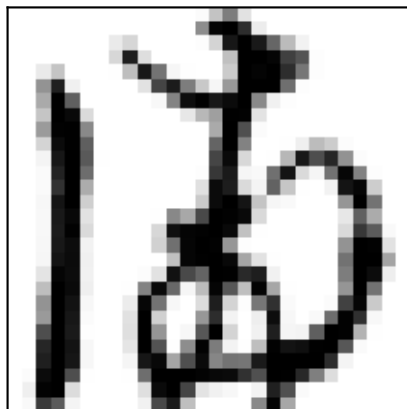


Diferença NEAT vs SGD

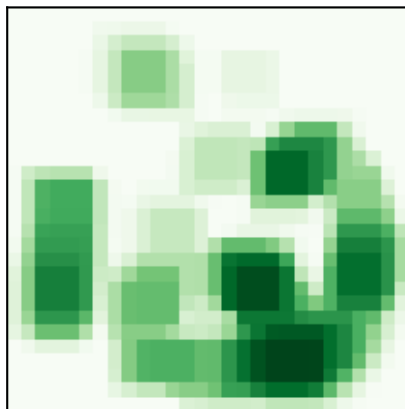


Baseline: 6 (51.0%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 6 (99.8%) | similaridade NEAT/SGD: 0.36 | pixels comuns: 35.0%

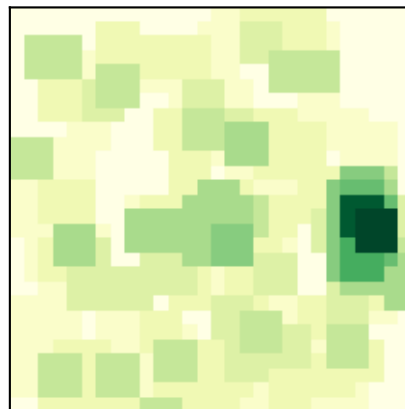
Imagem original



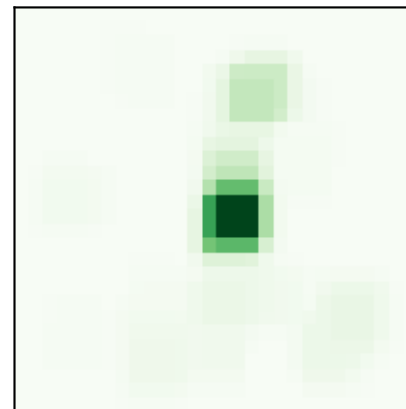
Olhar: baseline



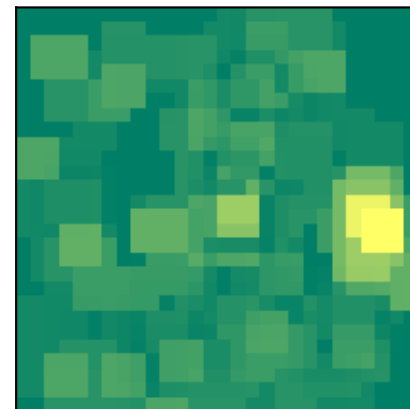
Olhar: NEAT puro



Olhar: NEAT + SGD

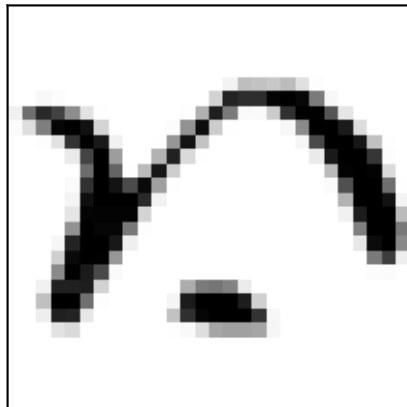


Diferenca NEAT vs SGD

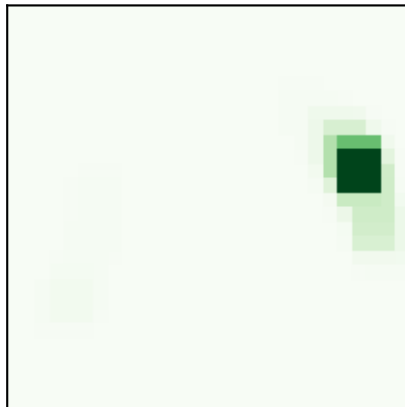


Baseline: 5 (100.0%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 5 (72.2%) | similaridade NEAT/SGD: 0.44 | pixels comuns: 37.6%

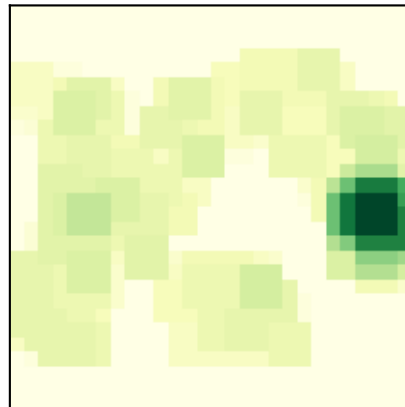
Imagem original



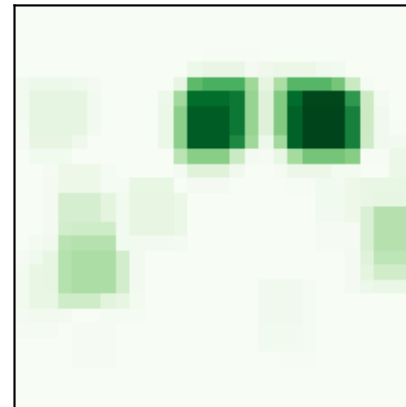
Olhar: baseline



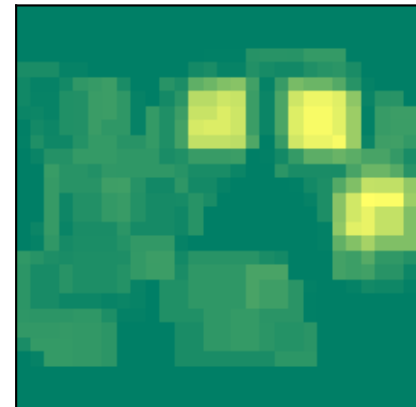
Olhar: NEAT puro



Olhar: NEAT + SGD

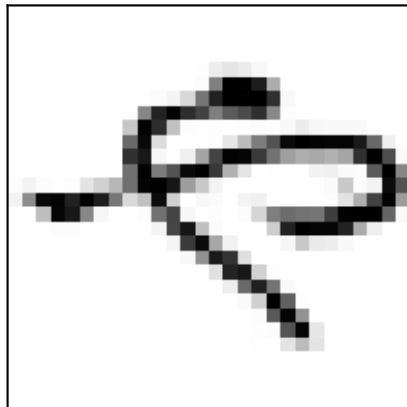


Diferença NEAT vs SGD

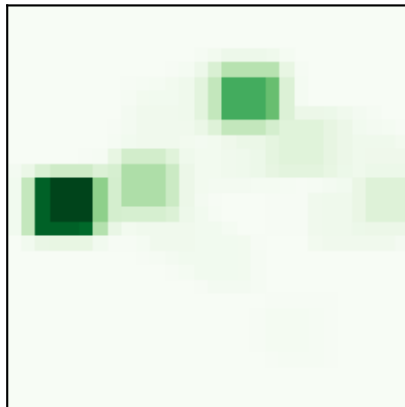


Baseline: 7 (100.0%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 7 (89.6%) | similaridade NEAT/SGD: 0.47 | pixels comuns: 42.7%

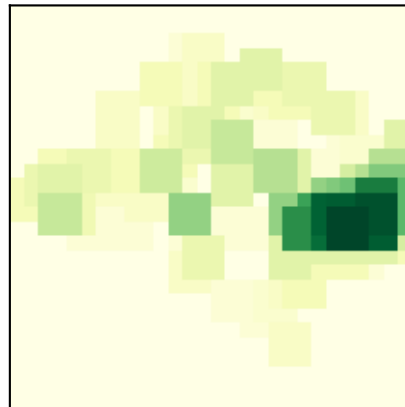
Imagem original



Olhar: baseline



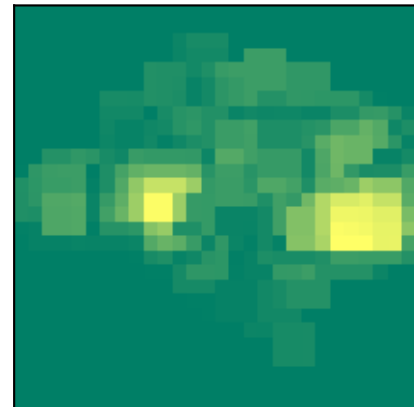
Olhar: NEAT puro



Olhar: NEAT + SGD

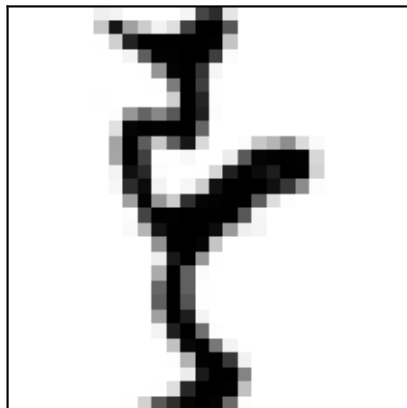


Diferença NEAT vs SGD

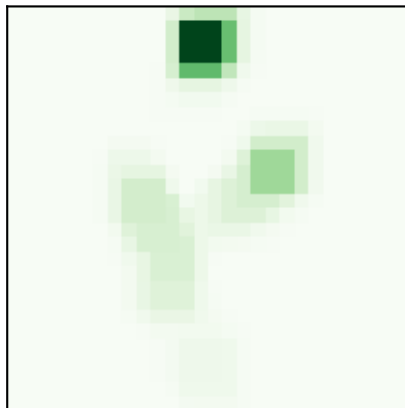


Baseline: 2 (99.8%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 2 (65.3%) | similaridade NEAT/SGD: 0.70 | pixels comuns: 41.0%

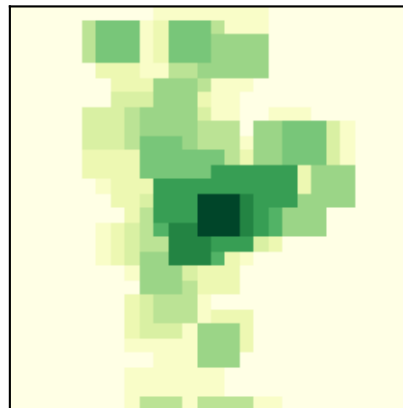
Imagem original



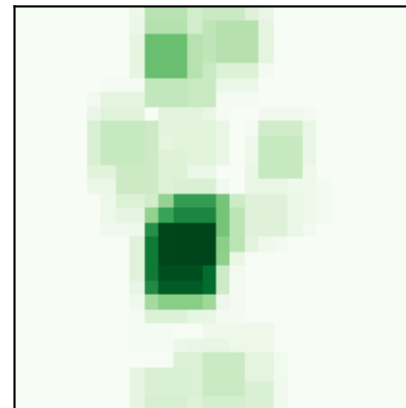
Olhar: baseline



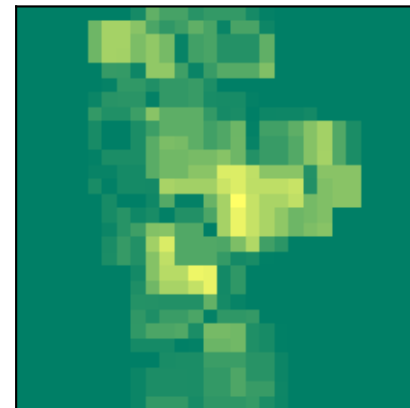
Olhar: NEAT puro



Olhar: NEAT + SGD



Diferenca NEAT vs SGD

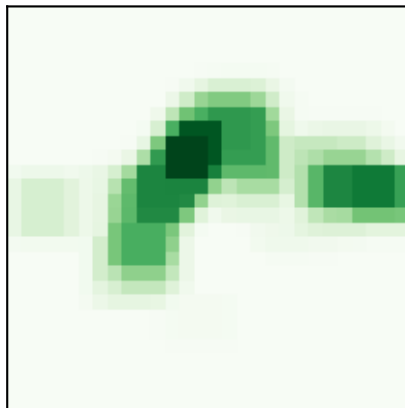


Baseline: 7 (99.9%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 7 (96.3%) | similaridade NEAT/SGD: 0.67 | pixels comuns: 38.5%

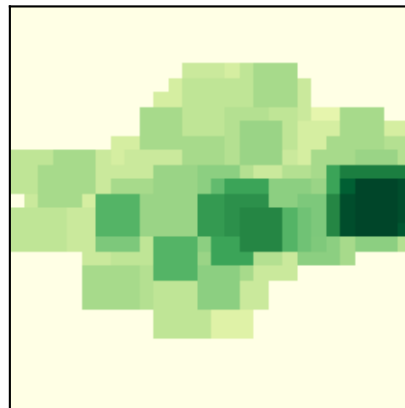
Imagem original



Olhar: baseline



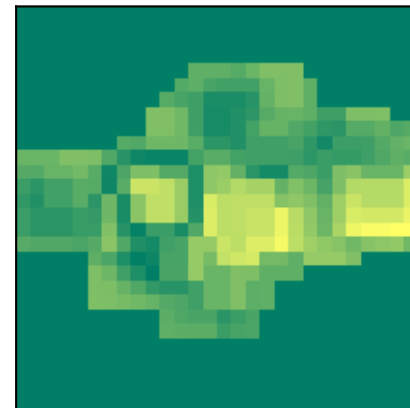
Olhar: NEAT puro



Olhar: NEAT + SGD



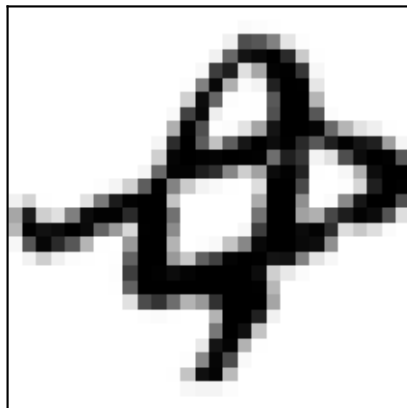
Diferença NEAT vs SGD





Baseline: 7 (100.0%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 7 (99.8%) | similaridade NEAT/SGD: 0.58 | pixels comuns: 35.0%

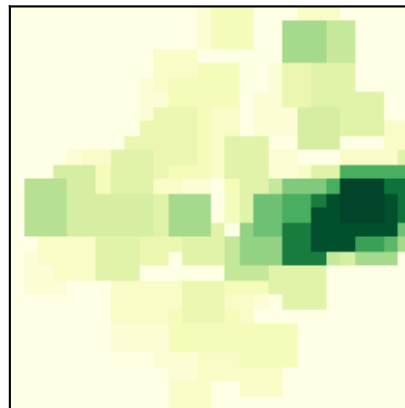
Imagem original



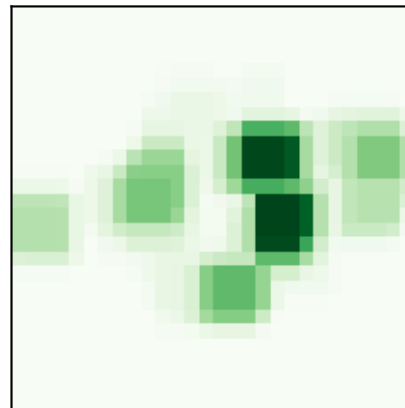
Olhar: baseline



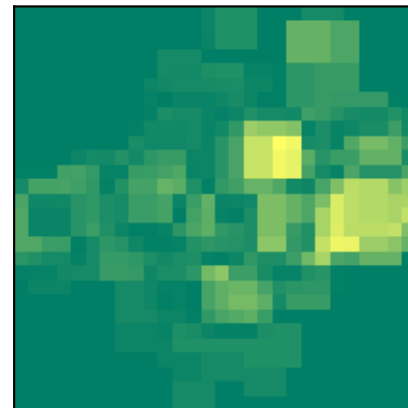
Olhar: NEAT puro



Olhar: NEAT + SGD

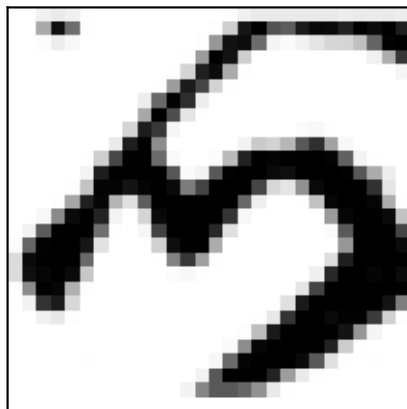


Diferença NEAT vs SGD

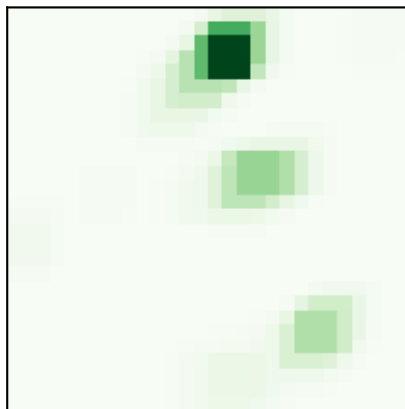


Baseline: 3 (100.0%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 3 (94.5%) | similaridade NEAT/SGD: 0.30 | pixels comuns: 18.8%

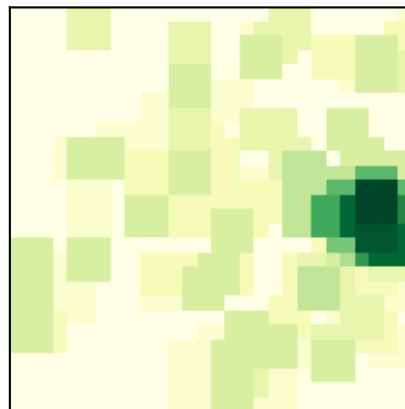
Imagem original



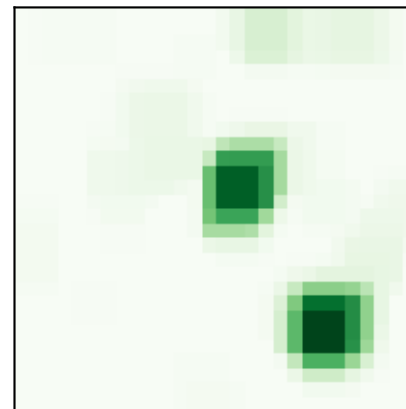
Olhar: baseline



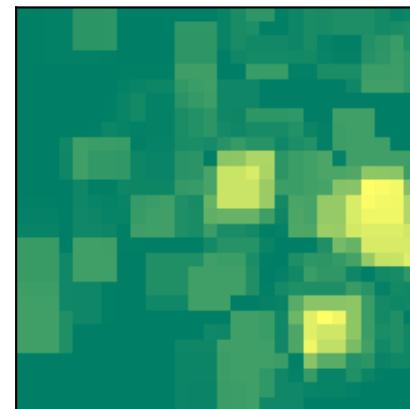
Olhar: NEAT puro



Olhar: NEAT + SGD

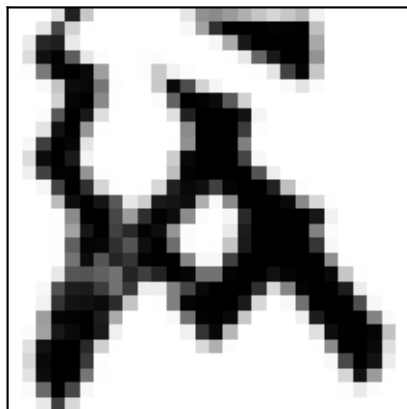


Diferença NEAT vs SGD



Baseline: 9 (100.0%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 9 (94.4%) | similaridade NEAT/SGD: 0.59 | pixels comuns: 35.9%

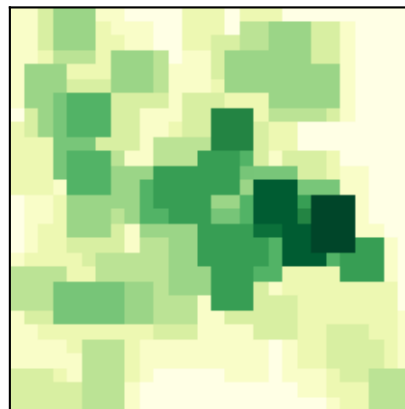
Imagem original



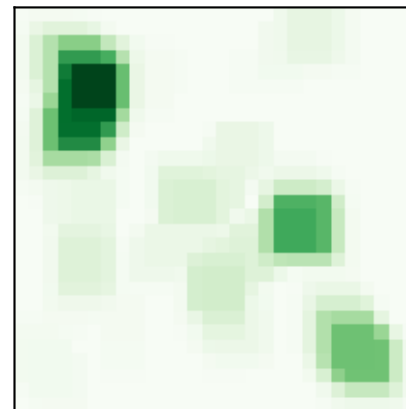
Olhar: baseline



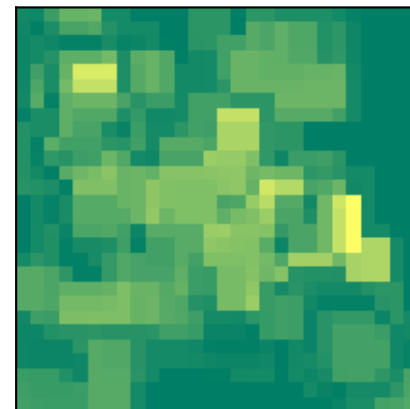
Olhar: NEAT puro



Olhar: NEAT + SGD

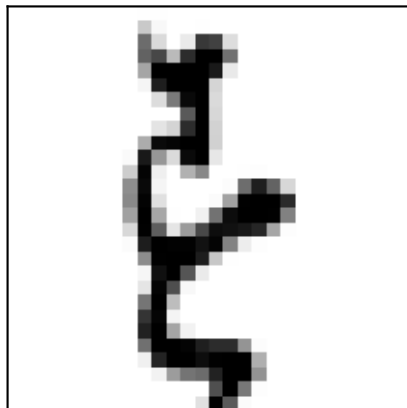


Diferença NEAT vs SGD

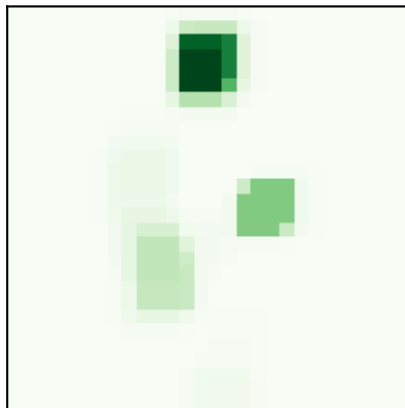


Baseline: 2 (99.5%) | NEAT puro: 0 (10.0%) | NEAT + SGD: 2 (78.1%) | similaridade NEAT/SGD: 0.74 | pixels comuns: 61.5%

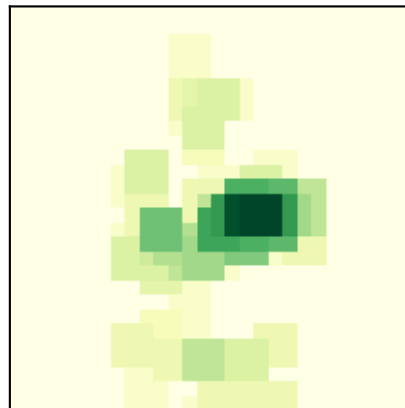
Imagem original



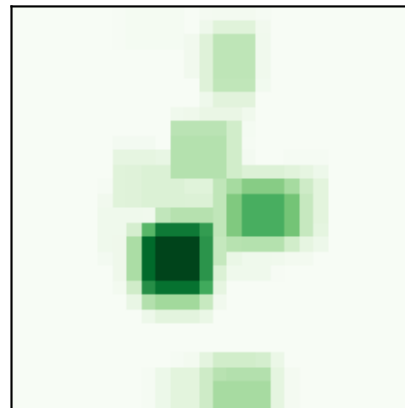
Olhar: baseline



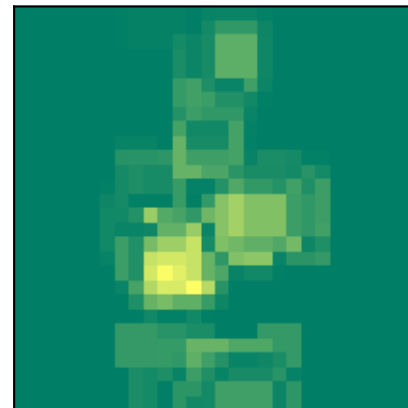
Olhar: NEAT puro



Olhar: NEAT + SGD

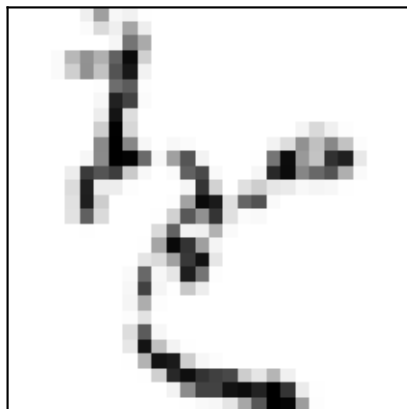


Diferença NEAT vs SGD

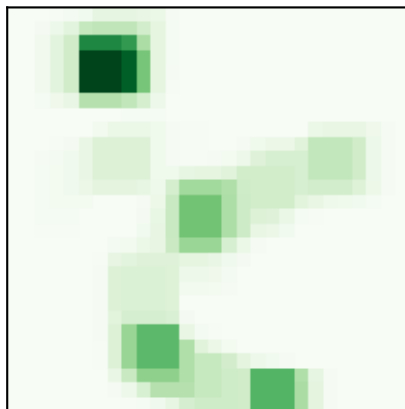


Baseline: 9 (100.0%) | NEAT puro: 1 (10.0%) | NEAT + SGD: 9 (55.4%) | similaridade NEAT/SGD: 0.68 | pixels comuns: 39.3%

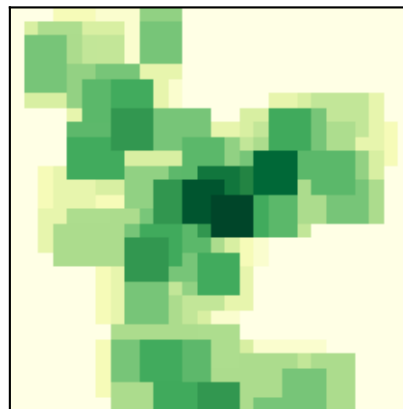
Imagem original



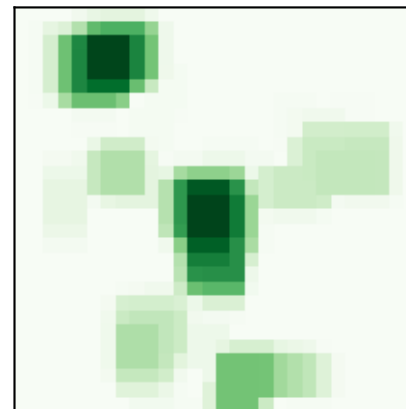
Olhar: baseline



Olhar: NEAT puro



Olhar: NEAT + SGD



Diferença NEAT vs SGD

